AIDAN HIGGS

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EDUCATION

University of Texas at Austin, Austin, TX 2026 – present
Ph.D. in Computational and Applied Mathematics

University of Florida, Gainesville, FL 2020 – 2024
B.S. in Mathematics and Biochemistry, GPA: 3.7/4.0

RESEARCH EXPERIENCE

National Institute of Mental Health/NIH & Princeton University Sep 2024 – Present
Mentor: Dr. Angela Langdon *NIH IRTA Postbac Fellow / Research Assistant*

Based at NIH campus in Bethesda, MD; initial appointment under a collaborative NIH U01 between Dr. Langdon and Professor Yael Niv.

- Developed and fit reinforcement learning models in Stan using Bayesian inference on rat choice behavior.
- Modeled electrophysiology data with generalized linear models incorporating latent variables from learning models.
- Integrated analyses of behavioral and fiber photometry data from mice in a reward-based task.
- Constructed and simulated variations of temporal difference learning agents.
- Implemented RNN architectures in Python/PyTorch with units exhibiting diverse intrinsic timescales.
- Analyzed how timescale heterogeneity affected training and dynamics of the trained network.

University of Florida, Department of Neuroscience Aug 2022 – Aug 2024
Mentor: Dr. Nancy Padilla-Coreano *Research Assistant*

- Trained SLEAP machine-learning models in Python to track mouse positions during behavioral assays.
- Conducted animal handling, behavioral testing, and managed food restriction protocols.
- Processed single-unit electrophysiological recordings; constructed custom electrodes for in vivo recordings.

University of Florida, Summer Neuroscience Internship Program May 2023 – Aug 2023
Mentor: Dr. Nancy Padilla-Coreano *Student Researcher*

- Assessed the validity of the tube test for determining dominance in CD-1 mice; examined reward effects on test outcomes.

Harvard T.H. Chan School of Public Health Aug 2021 – Dec 2021
Mentors: Dr. Kun Wang, Dr. Robert Farese & Dr. Tobias Walther *Research Assistant*

- Contributed to structural studies of MBOAT7, an enzyme in lipid metabolism.
- Generated mutant MBOAT7 genes for expression and structural analyses.
- Processed cryo-EM data using Bash scripts; analyzed lipidomic datasets with Python/Pandas.

PUBLICATIONS

Kim H. G., Gauthier M. L., **Higgs, A.**, et al. (2025). Chromatin Remodeling and Transcriptional Silencing Define the Dynamic Innate Immune Response of Tissue Resident Macrophages After Burn Injury. *Shock*. https://doi.org/10.1097/SHK.0000000000002746

Cum, M., Pérez, J. A. S., Iwata, R. L., Lopez, N., **Higgs, A.**, et al. (2024). A Multiparadigm Approach to Characterize Dominance Behaviors in CD1 and C57BL6 Male Mice. *eNeuro*. https://doi.org/10.1523/ENEURO.0342-24.2024

Wang, K., Lee, C., Sui, X., Kim, S., Wang, S., **Higgs, A.** et al. (2023). The structure, catalytic mechanism, and inhibitor identification of phosphatidylinositol remodeling MBOAT7. *Nature Communications*. https://doi.org/10.1038/s41467-023-38932-5

PRESENTATIONS

Recurrent neural networks with heterogenous time constants adapt efficiently to tasks with diverse timescales - Poster, Cold Spring Harbor Laboratory 90th Symposium on AI in Biology, Cold Spring Harbor, NY (2026)

Temporally-adaptive state representations facilitate flexible sequential prediction and learning — Invited talk, American Mathematical Society, Joint Mathematics Meetings, Special Session on Machine Learning in Mathematical Research and Applications, Washington D.C. (2026)

Recurrent Neural Networks with Diverse Intrinsic Timescales — Poster, NIMH training day, Bethesda, MD (2025)

Heterogeneous neural encoding of expected delay to and amount of reward in the VS during an odor-guided choice task — Poster, Multidisciplinary Conference on Reinforcement Learning and Decision Making, Dublin, Ireland (2025); NIH Postbac Poster Day, Bethesda, MD (2025)

Reward Modulation of Dominance Behavior: Examining the Tube Test in CD-1 Mice — Poster, Society for Social Neuroscience, Virtual (2024)

TEACHING EXPERIENCE

University of Florida, Department of Chemistry Fall 2023
Teaching assistant

Led exam review sessions, answered in-class questions, held office hours, and graded assignments for Biochemistry.

VOLUNTEERING

Norman Fixel Institute for Neurological Disease August 2023 – June 2024
Clinical volunteer

Helped a physician assistant coordinate information between patients, labs and pharmacies to ensure that the patients receive timely prescriptions for a highly regulated medication. Assisted nurses in taking patient vitals and rooming patients.

Lab of Dr. Takahiro Soda May 2023 – August 2023
Lab volunteer

Observed Dr. Soda discuss psychiatry cases in clinic. I then helped administer questionnaires to patients concerning their knowledge of genetic caused of autism spectrum disorder and the benefits of genetic testing.

AWARDS & HONORS

National Merit Scholar

Bright Futures Full Tuition Scholarship (2020 – 2024)

Florida Benacquisto Scholarship (2020 – 2024)

International Baccalaureate Diploma Recipient (2020)

TECHNICAL SKILLS

Programming Languages	Python, Bash, R, MATLAB, Stan, Julia
Machine Learning Tools	PyTorch, scikit-learn, Pandas, NumPy
Neuroscience Tools	SLEAP, GLMs, Ephys & Fiber photometry analysis